EXP 3

Aim

To design and simulate a 10 GHz microwave directional coupler with 10 dB coupling and a two-way 3 dB power divider, and analyze their S-parameters over 8–12 GHz.

Software required:

MATLAB

Program:

```
clc;
clear;
close all;

% Part A: Initial Design
f0 = 10e9;
f = linspace(8e9,12e9,401);
c = 3e8;
Z0 = 50;

% Directional coupler design
lambda0 = c/f0;
L = lambda0/4;
k = 10^(-10/20);

beta = 2*pi*f/c;
theta = beta*L;

% Simplified directional coupler
response
S11 = zeros(size(f));
S21 = cos(theta);
S31 = 1i*k.*sin(theta);
S41 = zeros(size(f));

% Convert to dB
S11_dB = 20*log10(abs(S11)+1e-
12);
S21_dB = 20*log10(abs(S21)+1e-
12);
S31_dB = 20*log10(abs(S31)+1e-
12);

% Plot directional coupler
figure;
plot(f/1e9,S11_dB,'k','LineWidth'
,1.5);
hold on;
plot(f/1e9,S21_dB,'b','LineWidth',1.5);
plot(f/1e9,S31_dB,'r','LineWidth',1.5);

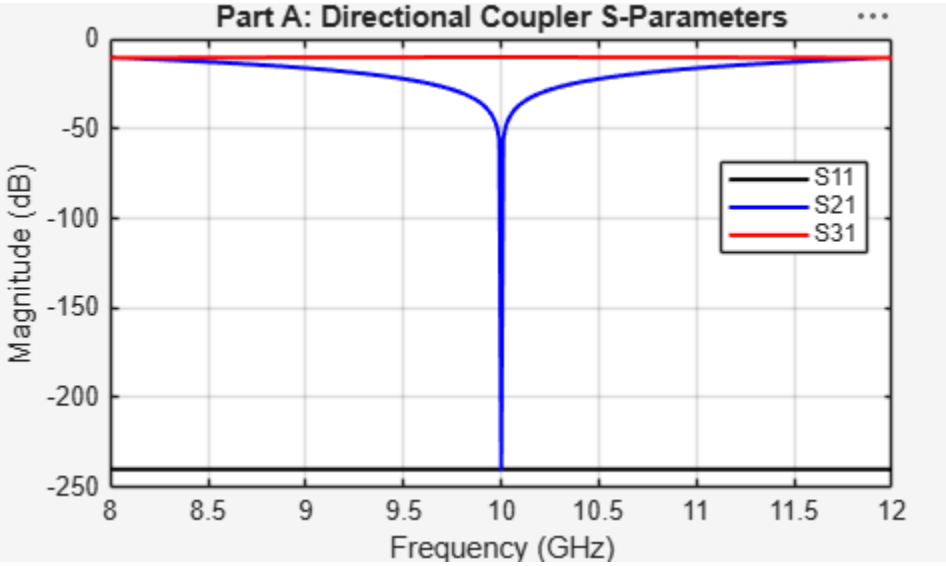
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Part A: Directional Coupler S-
Parameters');
legend('S11','S21','S31','Location','best');

% Two-way 3 dB power divider
S21_div = ones(size(f))/sqrt(2);
S31_div = ones(size(f))/sqrt(2);

figure;
plot(f/1e9,20*log10(abs(S21_div)),'b','LineWidt
h',1.5);
hold on;
plot(f/1e9,20*log10(abs(S31_div)),'r--
','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Part A: Two-Way 3 dB Power Divider');
legend('S21','S31','Location','best');

fprintf('Part A: Initial Design\n');
fprintf('Center frequency = %.2f
GHz\n',f0/1e9);
fprintf('Port impedance = %.0f Ohms\n',Z0);
fprintf('Quarter wavelength = %.2f
mm\n',L*1e3);
```

Output:




```
Part A: Initial Design
Center frequency = 10.00 GHz
Port impedance = 50 Ohms
Quarter wavelength = 7.50 mm
```

